

Customer Spending & Loyalty Analysis

A Python and MySQL portfolio project

Project question: What can a retailer learn about customer spend, product categories, loyalty, shipping preferences, and ratings from a 3,900-record shopping dataset?

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Tools: Python (pandas, matplotlib, seaborn) · MySQL · Jupyter Notebook

Dataset: Customer shopping behavior records supplied with the project brief

This report documents the analysis process, not only the final charts. All findings are descriptive because the file contains a static sample rather than a dated transaction history.

1. Project overview

The project studies customer-level shopping behavior across product categories, locations, customer demographics, ratings, subscription status, shipping types, discount use, and prior purchases. I used Python for cleaning and exploratory analysis, then wrote SQL questions that reproduce the main business views in a database environment.

What I wanted to answer

Business question	Why it matters
Which categories contribute the most recorded spend?	Helps focus category and inventory review.
Which customer groups generate the most spend?	Provides a starting point for retention or merchandising tests.
Do shipping and loyalty fields show meaningful differences?	Supports follow-up questions about customer experience.
Which product/season combinations deserve attention?	Adds context beyond overall category totals.

2. Dataset snapshot

Field group	Examples
Customer profile	Age, gender, location, previous purchases, purchase frequency
Purchase attributes	Item, category, price, size, color, season
Experience and loyalty	Review rating, shipping type, subscription status, discount applied
Checkout	Payment method, promo-code usage

The input file contains 3,900 rows and 18 source columns. There are 37 missing values in Review Rating; no full-row duplicate records were found.

3. Data preparation in Python

The notebook keeps the raw file unchanged and creates derived fields for analysis. Missing ratings are filled with the overall median rating (3.8) only for aggregation and plotting. This avoids losing records while making the treatment explicit.

Cleaning and feature engineering excerpt

```
clean = df.copy() clean['Review Rating'] = clean['Review Rating'].fillna(clean['Review Rating'].median()) clean['Age Group'] = pd.cut(clean['Age'], [0, 24, 34, 44, 54, 100], labels=['18-24', '25-34', '35-44', '45-54', '55+']) clean['Loyalty Group'] = pd.cut(clean['Previous Purchases'], [-1, 10, 30, 100], labels=['New (0-10)', 'Returning (11-30)', 'Loyal (31+)'])
```

Data checks

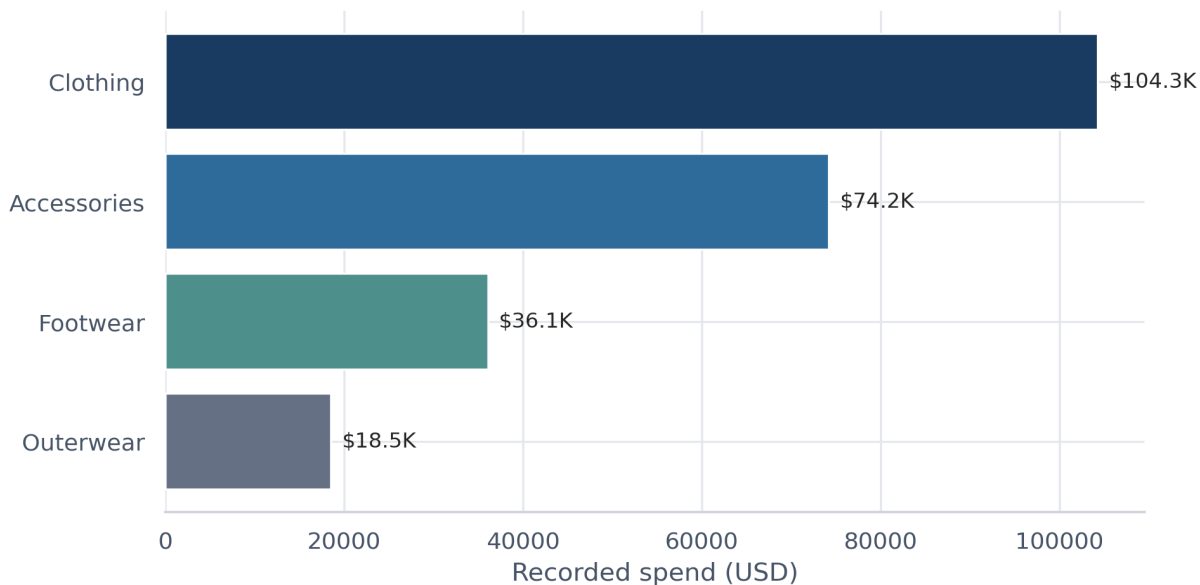
Check	Result	Decision
Row count	3,900	Retained all source records
Duplicate rows	0 full-row duplicates	No removal needed
Missing review ratings	37	Median imputation for summaries
Purchase amount	Numeric; \$20-\$100	Used directly in aggregations
Time dimension	Not available	No trend or retention-rate claim made

4. Portfolio baseline

Metric	Value
Recorded spend	\$233,081
Average purchase amount	\$59.76
Median review rating	3.8 / 5
Subscribers	1,053 (27.0%)
Discount users	1,677 (43.0%)

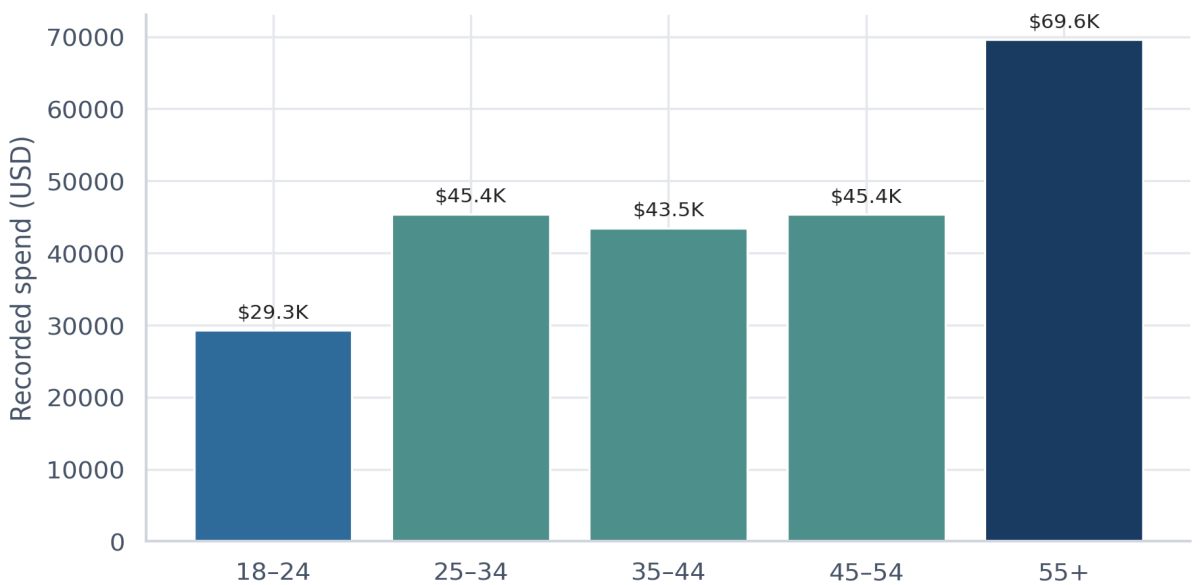
5. Category and customer findings

Clothing accounts for the largest share of recorded spend



Finding. Clothing contributes \$104.3K of the \$233.1K recorded spend (44.7%). Accessories is the next largest category at \$74.2K. These categories should be the first place to inspect product availability, customer feedback, and promotional depth.

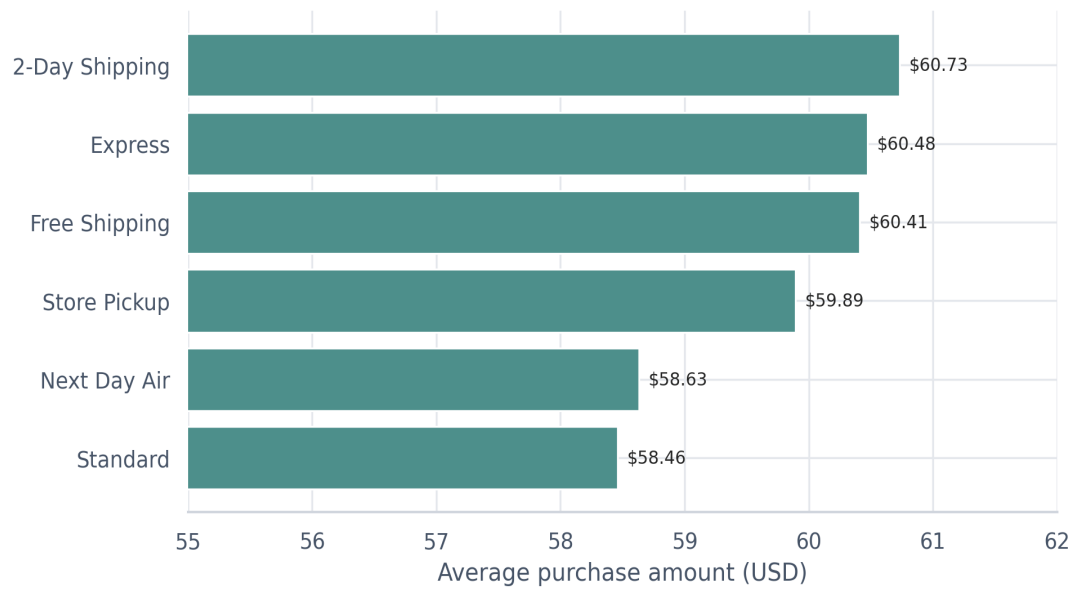
Customers aged 55+ contribute the largest total spend



Finding. Customers aged 55+ generate the largest total spend because this is also the largest age group in the dataset. Their average purchase value is not the highest, so the chart should not be read as proof that age drives order value.

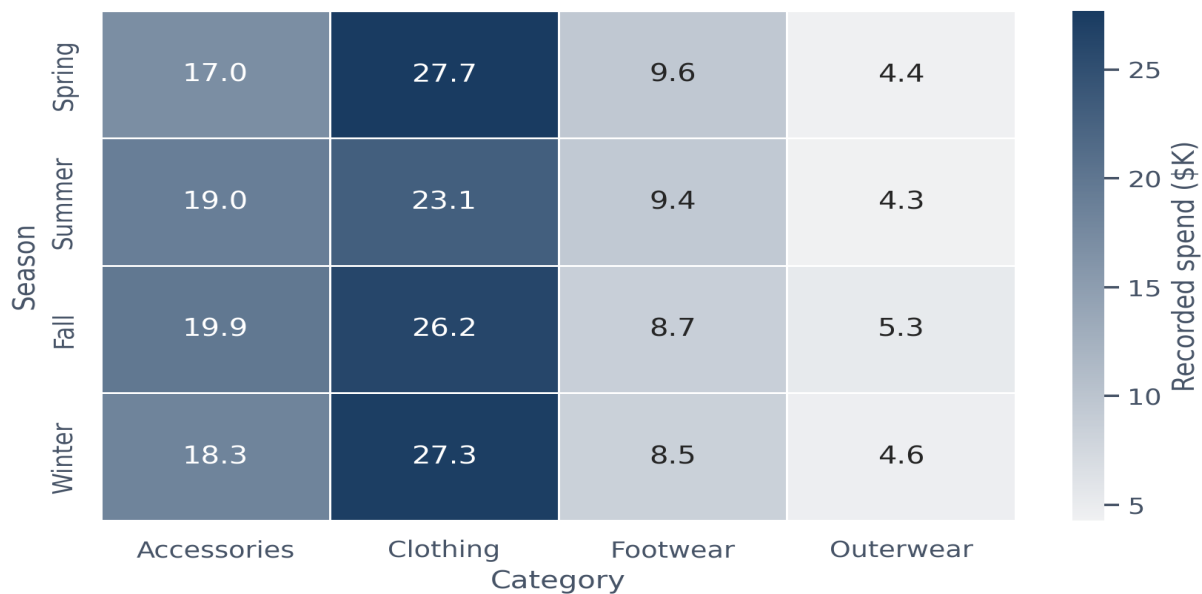
6. Shipping, seasons, and products

Average purchase varies by only \$2.27 across shipping methods



Finding. Average purchase values across shipping methods are tightly grouped. The observed range is \$58.46 to \$60.73, so the data does not support a strong conclusion that one delivery choice increases spending.

Clothing leads spend in every season



Finding. Clothing is the top category in every season. Fall has the highest recorded seasonal total (\$60.0K), but the data has no dates or external context to explain why.

7. SQL analysis questions

The accompanying customer_behavior_sql_queries.sql contains twelve queries. Below are examples of the questions that were translated from the exploratory analysis into SQL.

SQL question	Output
Portfolio overview	Customer count, recorded spend, average purchase, average rating, subscription
Category scorecard	Records, spend, average purchase, and rating by category
Segment comparison	Observed subscription/discount groups with counts, spend, prior purchases
Location ranking	Top locations by recorded spend
Product watchlist	Frequently purchased products with low average ratings
Frequency profile	Purchase frequency labels compared by spend and ratings

Example query: category scorecard

```
SELECT category, COUNT(*) AS customer_records, ROUND(SUM(purchase_amount_usd), 2) AS
recorded_spend_usd, ROUND(AVG(purchase_amount_usd), 2) AS average_purchase_usd,
ROUND(AVG(review_rating), 2) AS average_review_rating FROM retail_customers GROUP BY category ORDER BY
recorded_spend_usd DESC;
```

8. Recommendations

Recommendation	Reasoning	How to validate
Start category review with Clothing and Apparel. It has the largest recorded-spend total.	They represent the largest recorded-spend total.	Compare availability, margin, stock-outs, and product ratings.
Test—not assume—promotion segment averages are observational and close together.	Segment averages are observational and close together.	Use together to start group and measure incremental spend.
Use subscriber history for a retention pilot. Subscribers show higher average previous purchases.	Subscribers show higher average previous purchases.	Track repeat purchase, churn, and offer cost.
Add operational fields to future data. Current data lacks dates, cost, and inventory.	Current data lacks dates, cost, and inventory.	Combine transaction, product, and campaign data.

9. Limitations and learning notes

This project intentionally separates observation from proof. The data has one shopping record per customer ID and does not include a date field, product cost, store count, inventory position, campaign assignment, or a before/after outcome. It is appropriate for descriptive analysis and generating hypotheses, but not for estimating sales lift, customer lifetime value, or causal promotion effects.

What I learned during the project

- Standardizing and checking data before visualization prevents misleading summaries.
- A chart title should communicate the main takeaway, not merely restate the fields.
- SQL and Python can answer the same business question in complementary ways: Python supports exploration and plotting, while SQL makes the analysis reusable in a database.
- Stating limitations makes recommendations more credible.

Repository contents

Asset	Purpose
Customer_Shopping_Behavior_Analysis.ipynb	Full exploratory workflow with executed outputs
analysis.py	Quick reproducible script
customer_behavior_sql_queries.sql	Twelve documented MySQL questions
Business Problem Document.pdf	Scope, stakeholders, and intended use
Customer Shopping Behavior Analysis.pdf	This report
Customer-Shopping-Behavior-Analysis.pptx	Presentation summary
assets/ and CSV summaries	Original charts and reusable outputs

Appendix: chart design note

Bar charts are used for category and shipping comparisons because they allow accurate comparison across discrete groups. The seasonal matrix is a heatmap because it shows the relationship between two categorical fields. Labels, units, and takeaway titles are included on every chart so figures can be read without surrounding text.